EEP 564: TinyML

Lab 7: TinyML for Motion Classification – On Device Data Collection & Ensemble Learning

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ELECTRICAL & COMPUTER ENGINEERING
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Today's Lab

Part I (Edge Impulse): On Device Data Collection and Motion Classification

- Both Software and Hardware Lab!
- Accelerometer Data Collection for Motion Classification
- Computing FFT, Spectrogram, and Power Spectral Density (PSD)
- Feature Extraction from Time Series IMU Data
- Training ML Models for Motion Classification
- Implementing the Model on Arduino Nano 33 BLE Sense using TinyML

Lab 7 – Part I - Software - Road Map

1. Installing Arduino CLI

2. Installing Edge Impulse Firmware on Arduino

3. Data Collection and Labelling

4. Splitting Data

5. Spectral Analysis

6. Training the ML model (DNN)

Lab 7 – Part I - Software - Road Map

7. Evaluation of ML Model Performance

8. Building ML model

Lab 7 – Part I - Software Lab – Arduino CLI Installation

If you're on Windows:

- Unzip the .zip file to C:\Program Files\arduino-cli
- Open System Properties > Advanced > Environment Variables
- Path under "user variables" > Edit
- Add C:\Program Files\arduino-cli

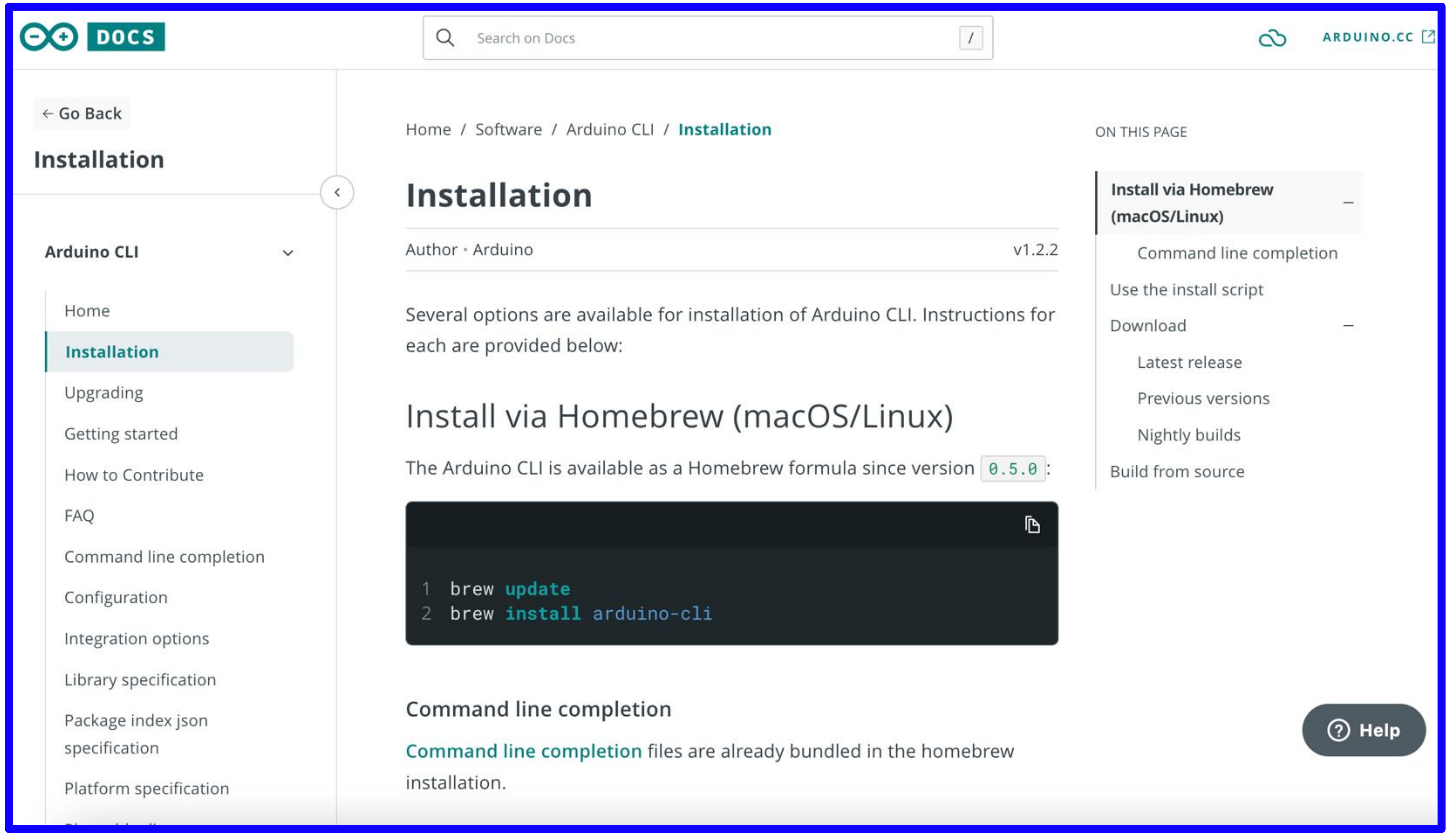
You can skip this step if you have already installed Arduino CLI in Lab 5 or Lab 6

If you're on macOS:

- Use the curl method or homebrew shown on the installation page: https://arduino.github.io/arduino-cli/0.21/installation/
- If you use the curl method, it likely installs arduino-cli to ~/bin. That might not be on your path. So, you might need to run: export PATH=$PATH:~/bin

Lab 7 – Part I – Software Lab: Setting up Arduino CLI

- Go to https://docs.arduino.cc/arduino-cli/installation/
- Carefully follow instructions corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for Download and Install Arduino CLI



You can skip this step if you have already installed Arduino CLI in Lab 5 or Lab 6

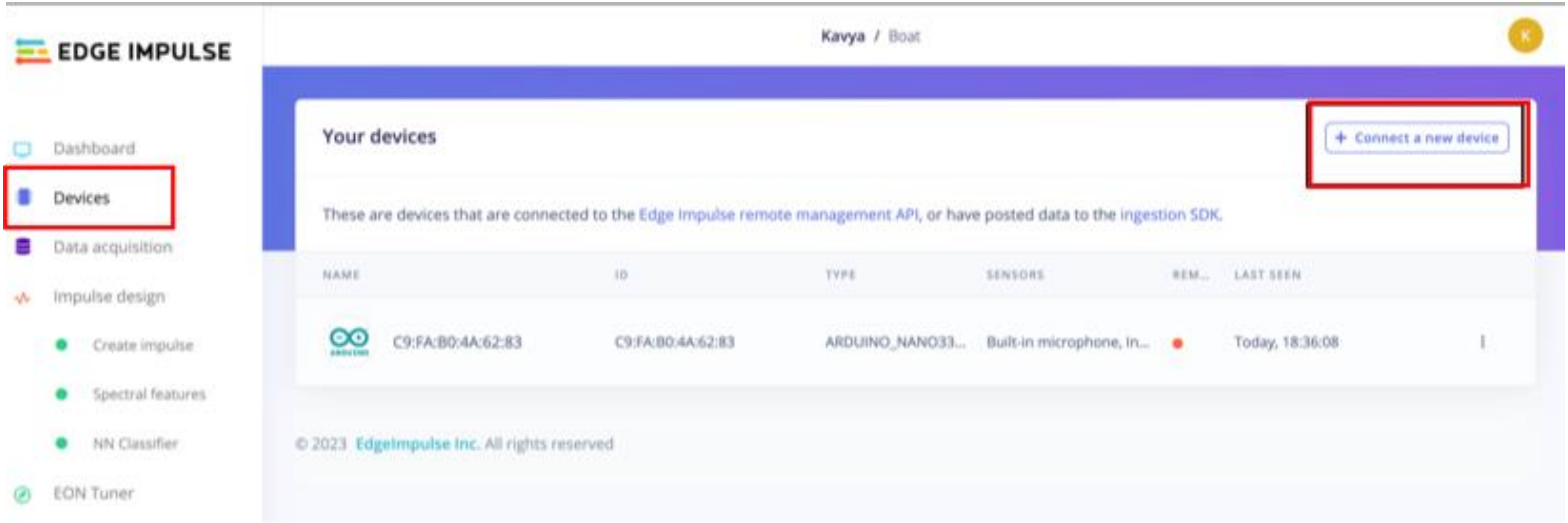
Lab 7 – Part I – Software Lab – Edge Impulse Project Creation

- We will use Edge Impulse for this Lab.
  Website: https://www.edgeimpulse.com/
  (You can login in using the account we created for the previous Edge Impulse lab)

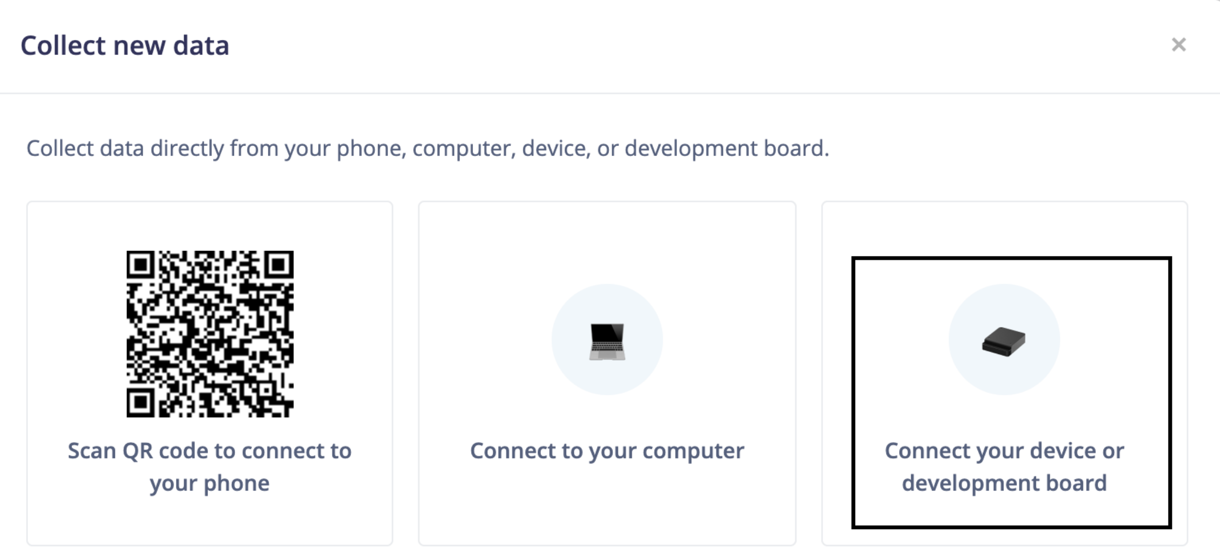
- Create a new Project on Edge Impulse and provide the title as "Motion Classification for Transport".

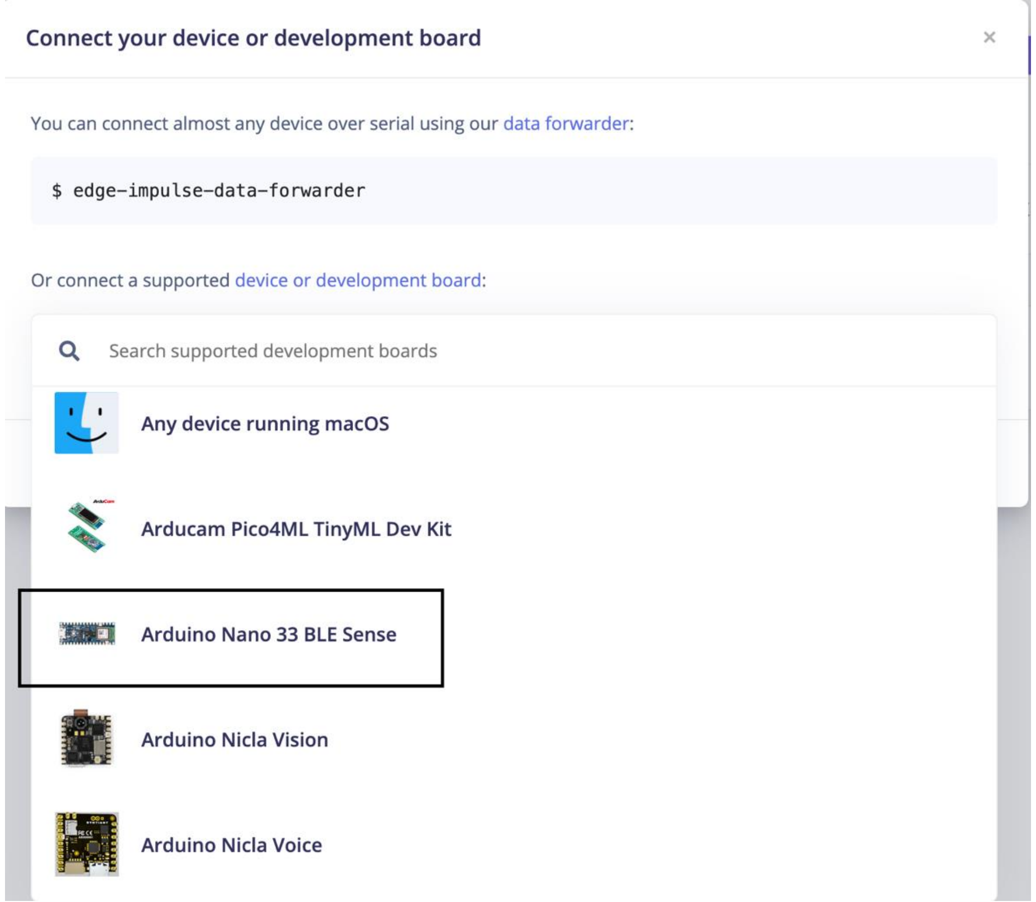
Software Lab – Connecting the Device

- First connect your Arduino Nano BLE to your Laptop/PC using USB C Cable
- Then in your Edge Impulse Project go to “Devices”

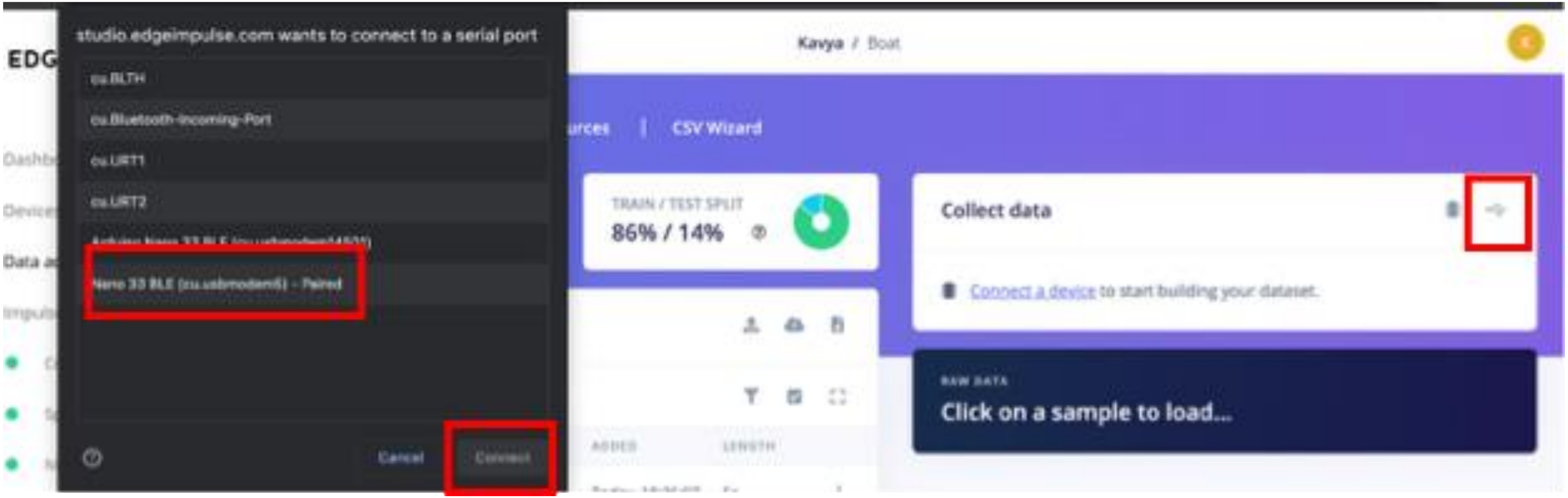


Software Lab - Connecting the Device

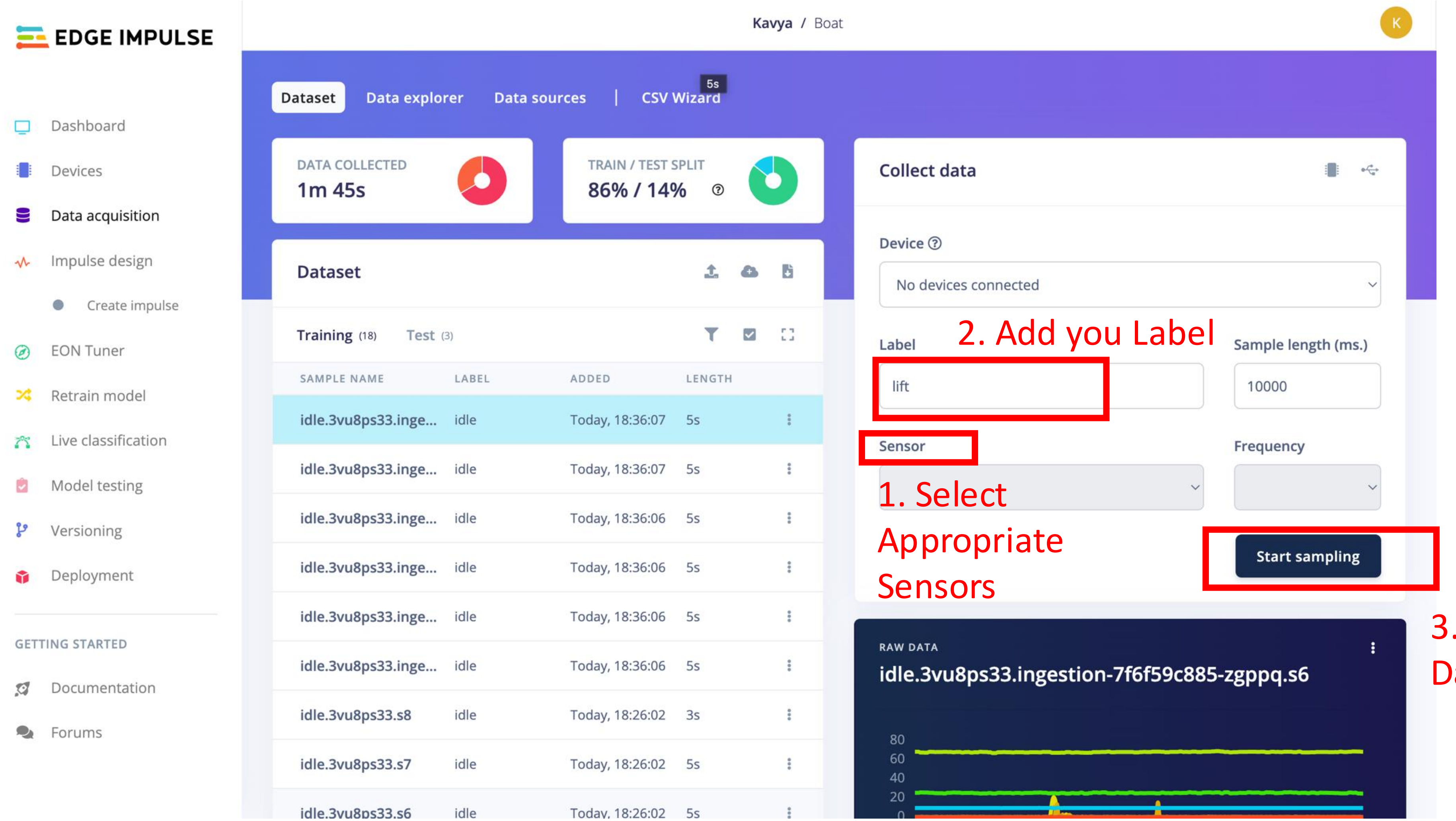




Software Lab – Data Acquisition



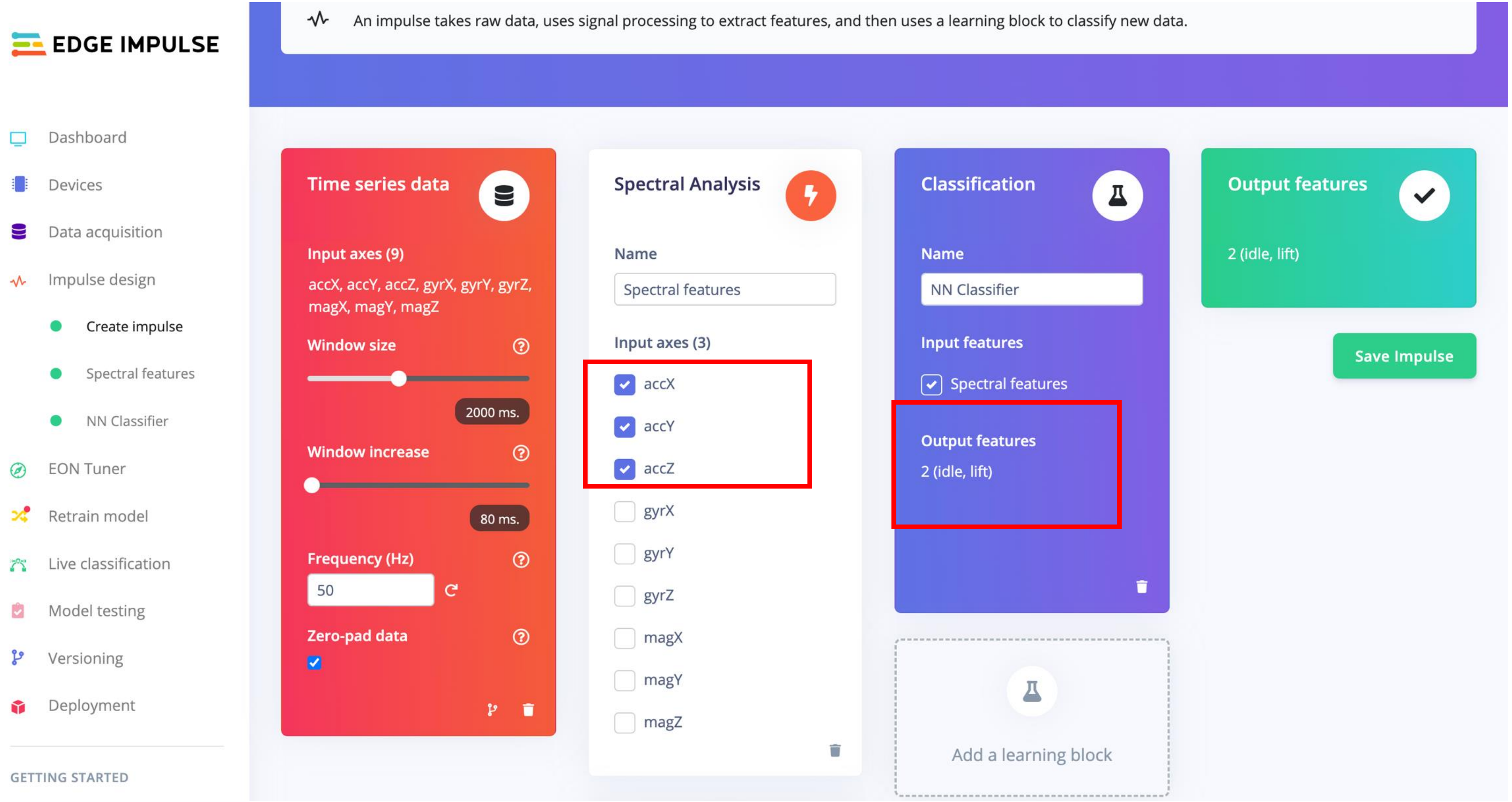
Software Lab – Data Acquisition



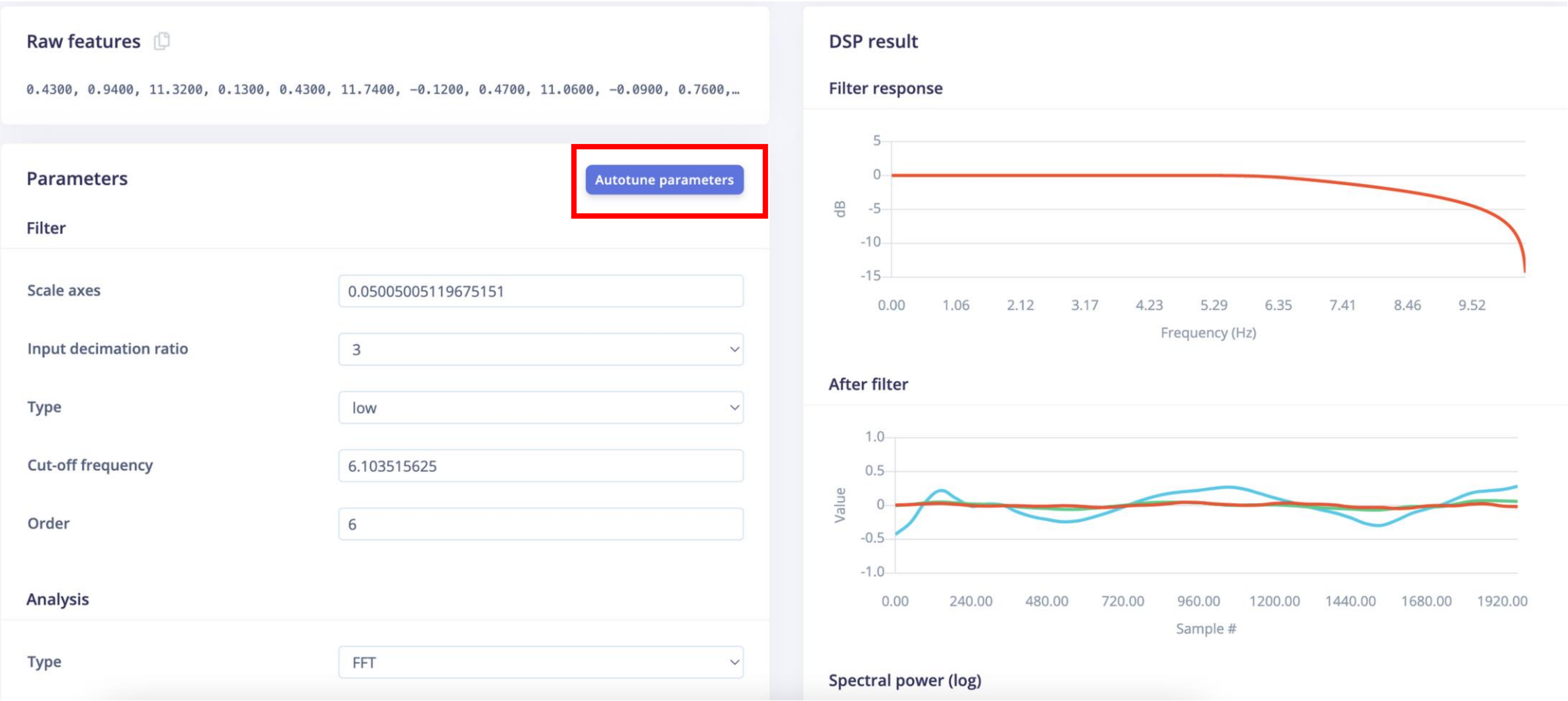
For Lab 7, create 20 samples per class—'Lift' and 'Idle'—each sample lasting 10 seconds.

3. Press to Collect Data point

Software Lab – Impulse Design – Create Impulse

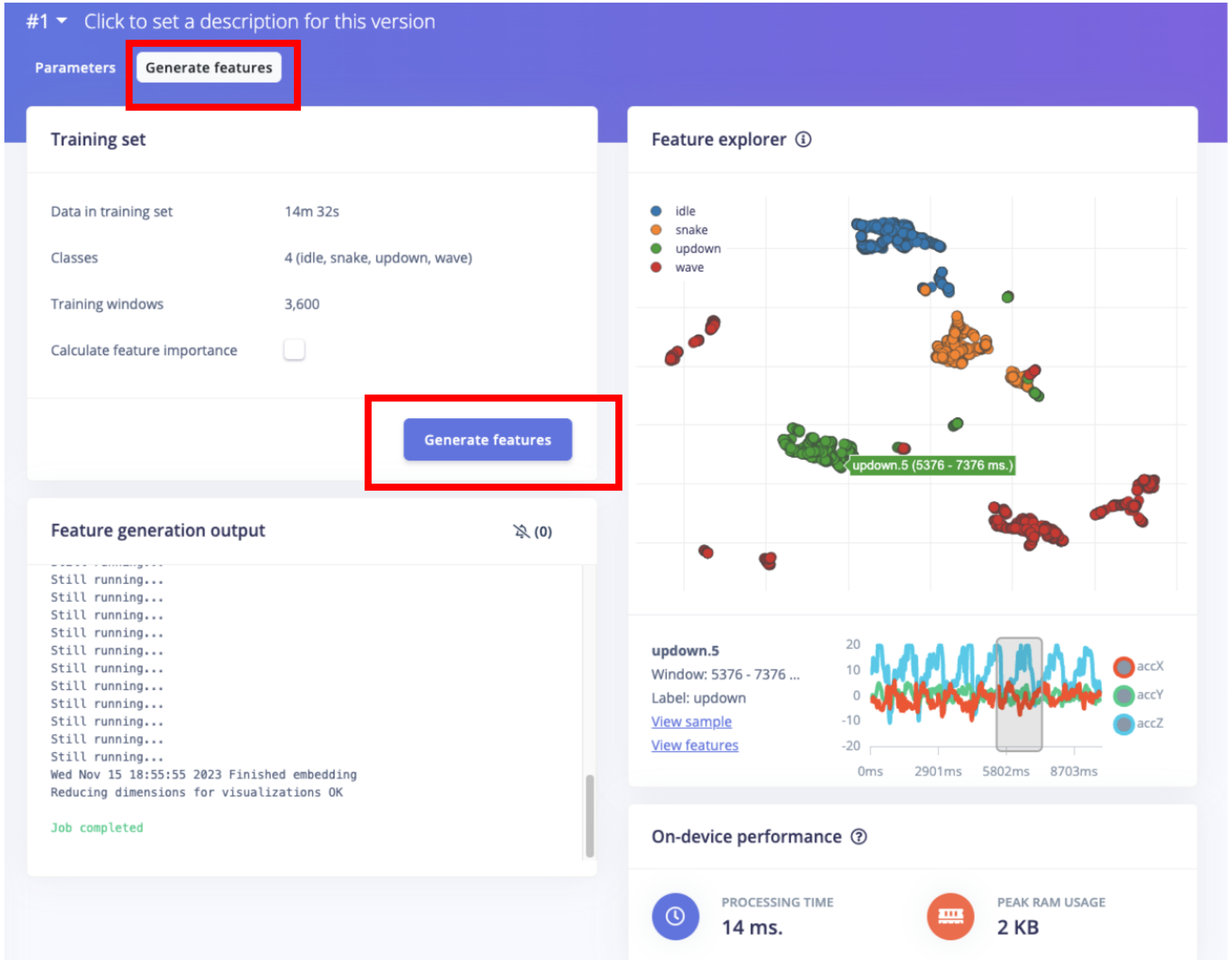


Software Lab – Impulse Design – Spectral Features

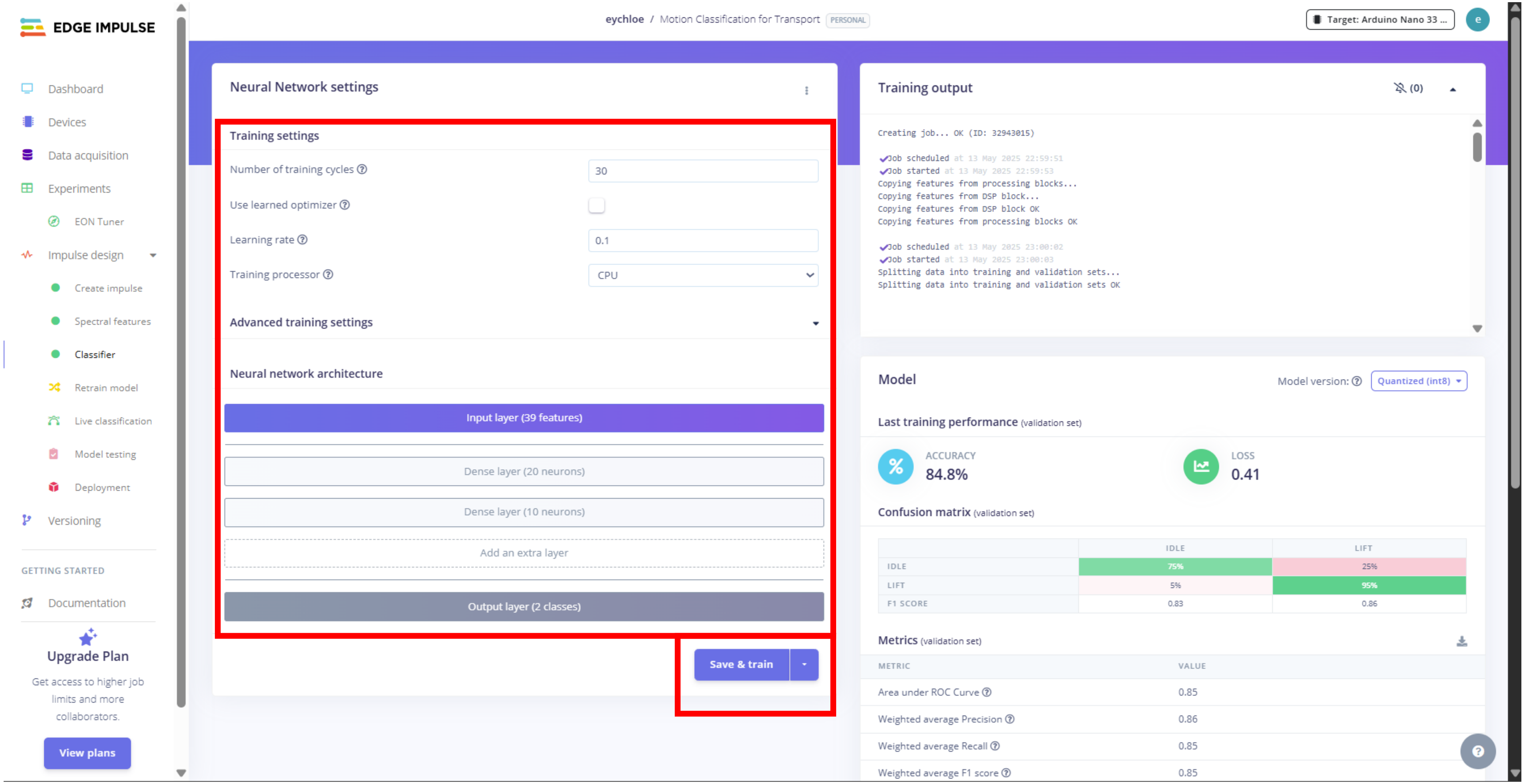


https://docs.edgeimpulse.com/docs/edge-impulse-studio/processing-blocks/spectral-features

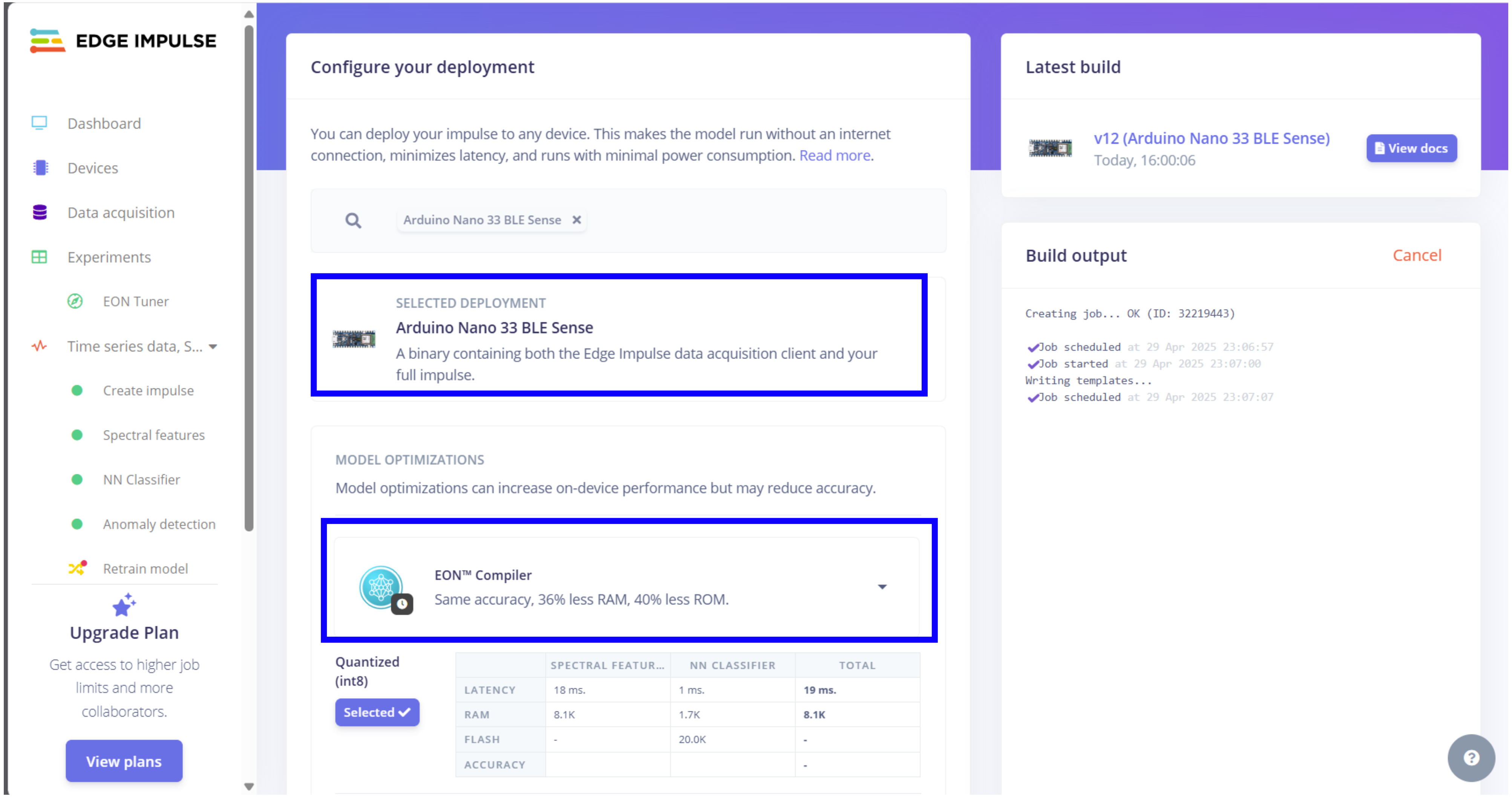
Software Lab – Impulse Design – Spectral Features



Software Lab – Impulse Design - NN Classifier

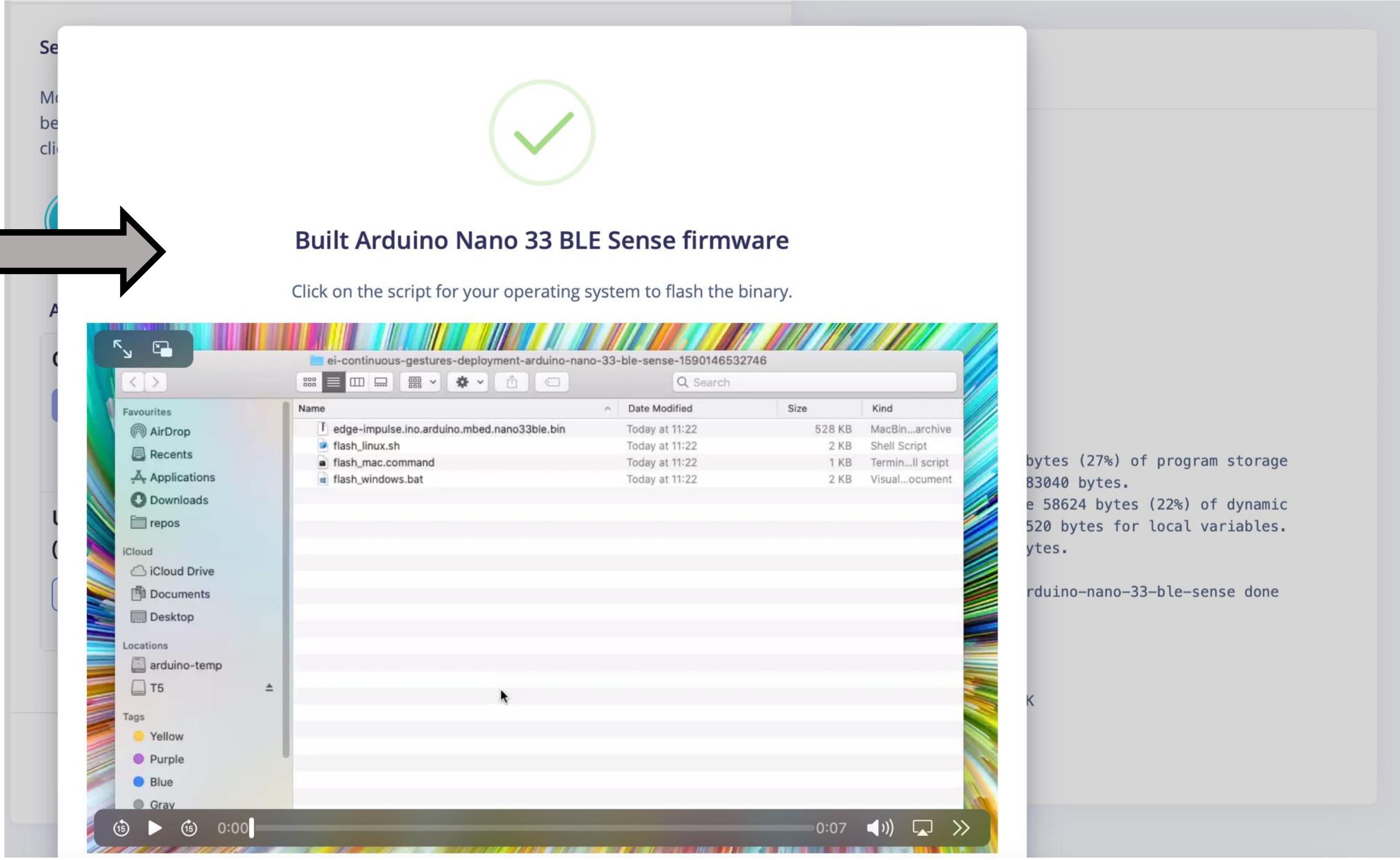


Software Lab – Deployment



Software Lab – Model Compression

Upon a Successful Model Build You Should Be Able to See Something Like This



Lab 7 – Part I - Hardware - Road Map

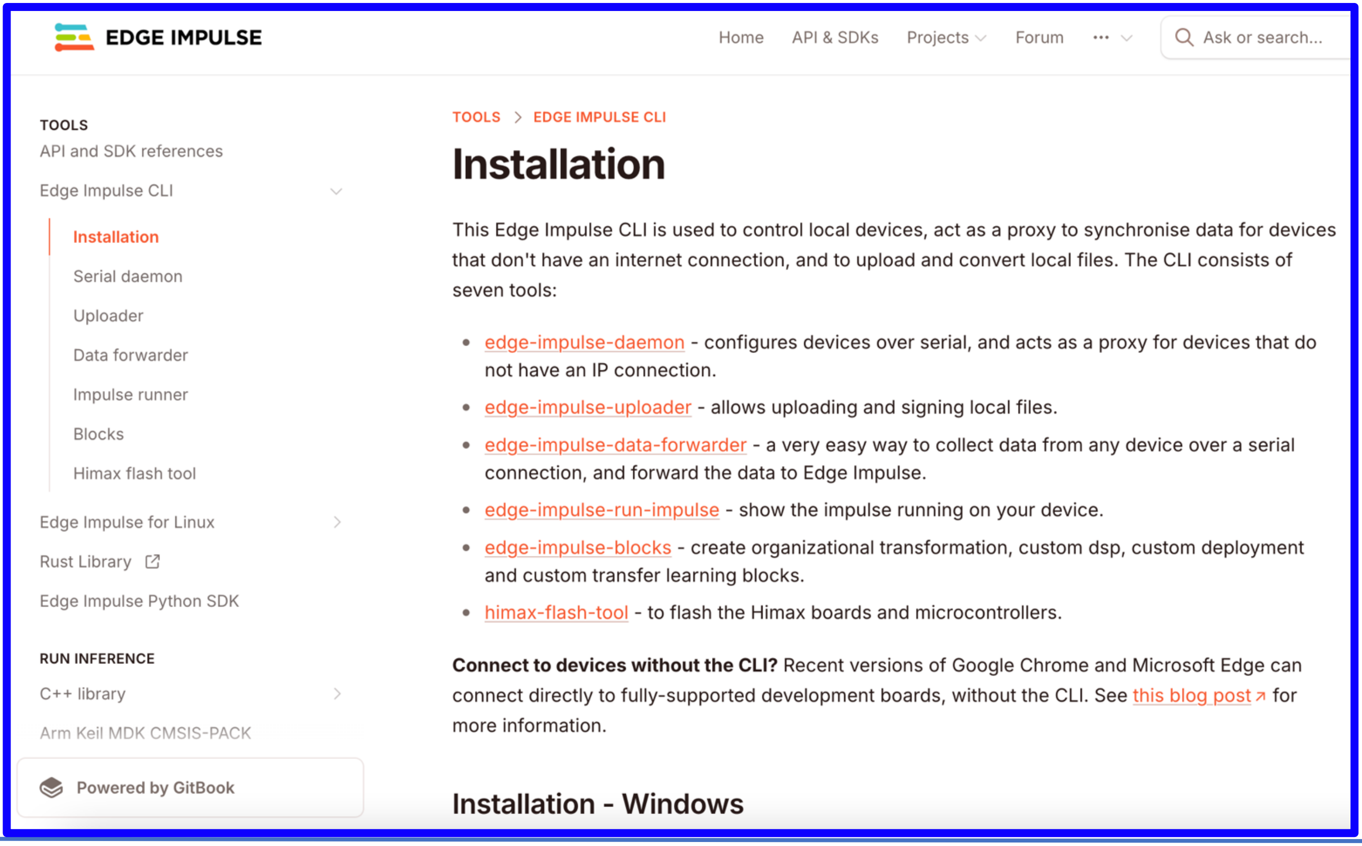
1. Build the Arduino Library/Flash files using Edge Impulse.

2. Compile and Upload .zip/Flash the sketch to the board.

3. Open Serial Monitor/Command Prompt and test the classification.

Hardware Lab: Setting up Edge Impulse CLI

- Go to https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation
- Carefully follow instructions corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for Download and Edge Impulse CLI


EDGE IMPULSE

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TOOLS
API and SDK references
Edge Impulse CLI
Installation
Serial daemon
Uploader
Data forwarder
Impulse runner
Blocks
Himax flash tool
Edge Impulse for Linux
Rust Library
Edge Impulse Python SDK

RUN INFERENCE
C++ library
Arm Keil MDK CMSIS-PACK

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TOOLS > EDGE IMPULSE CLI

Installation

This Edge Impulse CLI is used to control local devices, act as a proxy to synchronise data for devices that don't have an internet connection, and to upload and convert local files. The CLI consists of seven tools:

- edge-impulse-daemon - configures devices over serial, and acts as a proxy for devices that do not have an IP connection.
- edge-impulse-uploader - allows uploading and signing local files.
- edge-impulse-data-forwarder - a very easy way to collect data from any device over a serial connection, and forward the data to Edge Impulse.
- edge-impulse-run-impulse - show the impulse running on your device.
- edge-impulse-blocks - create organizational transformation, custom dsp, custom deployment and custom transfer learning blocks.
- himax-flash-tool - to flash the Himax boards and microcontrollers.

Connect to devices without the CLI? Recent versions of Google Chrome and Microsoft Edge can connect directly to fully-supported development boards, without the CLI. See this blog post for more information.

Installation - Windows

You can skip this step if you have already installed Edge Impulse CLI in Lab 5

Hardware Lab – Arduino Deployment & Testing

- Flash the compressed model (downloaded from Edge Impulse) onto the board by double clicking on the script for your operating system

- Follow the instructions prompted in command line to observe the results in command line (*you may need to type the prompted commands in a new command prompt window*)

```
/Users/Admin/Downloads/arduino-nano-33-ble-sense\ \(2\)/flash_mac.command ; exit;
(base) Admin@Sarats-MacBook-Pro ~ % /Users/Admin/Downloads/arduino-nano-33-ble-sense\ \(2\)/flash_mac.command ; exit
You're using an untested version of Arduino CLI, this might cause issues (found: 0.32.2, expected: 0.18.x)
Finding Arduino Mbed core...
Finding Arduino Mbed OK
Finding Arduino Nano 33 BLE...
Finding Arduino Nano 33 BLE OK
Flashing board...
TOUCH: error during reset: opening port at 1200bps: Serial port busy
Device       : nRF52840-QIAA
Version      : Arduino Bootloader (SAM-BA extended) 2.0 [Arduino:IKXYZ]
Address      : 0x0
Pages        : 256
Page Size    : 4096 bytes
Total Size   : 1024KB
Planes       : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.001 seconds
Write 352560 bytes to flash (87 pages)
[==============================] 100% (87/87 pages)
Done in 13.914 seconds

Flashed your Arduino Nano 33 BLE development board.
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

Today’s Lab

Part II (Google Collab + Arduino IDE): Motion Classification and Tiny Ensemble Learning

- Both Software and Hardware Lab!
- Model Ensemble Techniques for Improved Accuracy
- Compressing Models using Pruning and Quantization-Aware Training (QAT)
- Combining Multiple Classifiers with a Stacked Meta-Classifier
- Real-World Testing with Recorded IMU Data
- Deploying the Final Model on Arduino Nano 33 BLE Sense using TFLite

Lab 7 – Part II – Tiny Ensemble Learning

🗁 Open and run **TinyML-Lab7-Part-II.ipynb**

📝 Carefully follow the instructions and **answer the discussion questions given in TinyML-Lab7-Part-II.ipynb**

📄 Submit **one PDF** with answers to **Part II Questions of Lab 7**

📄 You are also **encouraged to submit Output of Part I of Lab 7**, though Part I is optional